INSTRUCTORMANUAL

CSE 4571 —PRACTICAL ROBOTICS PROJECTS WITH ARDUINO

Seventh Semester B.Tech (2L–4P)| Hardware-Integrated Simulation-Based Laboratory Practical

Arduino Robotics — Lukas Kaul

SESSION- 5

SENSOR-BASED AUTOMATIC ACTUATOR CONTROL SYSTEM

Learning Domain Distribution—Cognitive:45% |Psychomotor:40% |Affective:15%

Note: This is a Hardware-Integrated Simulation-Based Laboratory course, where simulation activities are supported by actual hardware components. Physical components, breadboards, and hardware kits are required for laboratory activities, along with the prescribed simulation

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1. Session Information

Field	Details
Course Code/Title	CSE 4571 — Practical Robotics Project with Arduino
Semester	Seventh Semester B.Tech (CSE)
Course Format	2L – 4P (Practical / Laboratory, Hardware-Integrated and Simulation-Based)
Session	5
Duration	6 Hours (360 minutes)
Software Used	Tinkercad / Wokwi (Online Circuit Simulation Platform) and physical components
Topic	Sensor-Based Automatic Actuator Control System Using Arduino
Text book Reference	Practical Arduino Robotics — Lukas Kaul

2. Learning Objectives

By the end of this session, students will be able to:

Objectives	Statements
O1	Explain the fundamental concept of sensor-based automatic actuator control systems and their relevance to modern engineering applications.
O2	Differentiate between various sensors and actuators based on their functions and applications in an automatic control system.
O3	State the role of sensors in detecting physical parameters and providing input signals for automatic actuator control.
O4	Navigate the simulation environment and identify the required sensors, actuators, microcontroller, power supply, and connecting components.
O5	Demonstrate the interfacing of a sensor with an actuator and simulate an automatic control system based on predefined sensor conditions.

3. Course Outcomes (CO1–CO6)

CO	Course Outcome Statement	Learning Domain	Taxonomy Level
CO1	Explain the fundamentals of Arduino-based robotic systems and interface different sensors for measurement, analog data acquisition, and real-time sensing applications.	Cognitive (L) + Psychomotor (P) + Affective (A)	Cognitive (L) + Psychomotor (P) + Affective (A)
CO2	Develop Arduino programs to control robotic actuators including servo motors, stepper motors, and DC motors for position and motion-based applications.	Cognitive (L) + Psychomotor (P) + Affective (A)	Cognitive (L) + Psychomotor (P) + Affective (A)
CO3	Implement intelligent robotic control systems using sensor inputs, finite state machines, non-blocking programming techniques, and feedback-based control mechanisms.	Cognitive (L) + Psychomotor (P) + Affective (A)	Cognitive (L) + Psychomotor (P) + Affective (A)
CO4	Design modular and reusable robotic software components using object-oriented programming	Cognitive (L) + Psychomotor (P) +	Cognitive (L) + Psychomotor (P) +

	concepts and Arduino libraries for embedded applications.	Affective (A)	Affective (A)
CO5	Analyze and troubleshoot robotic systems by integrating communication interfaces, remote-control techniques, and wireless monitoring using embedded platforms.	Cognitive (L) + Psychomotor (P) + Affective (A)	Cognitive (L) + Psychomotor (P) + Affective (A)
CO6	Design and demonstrate an Arduino-based robotic application by integrating sensors, actuators, control algorithms, and wireless communication modules while following engineering practices.	Cognitive (L) + Psychomotor (P) + Affective (A)	Cognitive (L) + Psychomotor (P) + Affective (A)

L:Cognitive(L1:Remembering;L2:Understanding;L3:Applying;L4:Analyzing;L5:Evaluating;L6:Creating)

P: Psychomotor (P1: Perception; P2: Set; P3: Guided Response; P4: Mechanism; P5: Complex Overt Response; P6: Adaptation; P7: Origination)

A:Affective(A1:Receiving;A2:Responding;A3:Valuing;A4:Organization;A5:Characterization)

The course CSE 4571– Practical Robotics Projects with Arduino primarily reflects the following Sustainable Development Goals (SDGs):

SDG	Goal	Course Relevance
SDG4	Quality Education	Develops practical skills in Arduino programming, robotics, electronics, automation, and problem-solving through hands-on projects.
SDG 9	**Industry, Innovation and Infrastructure**	Promotes innovation, technological development, automation, and the practical application of robotics and embedded systems.
SDG 11	Sustainable Cities and Communities	Encourages the development of smart robotic and automated solutions for safer, more efficient, and sustainable communities.
SDG 12	Responsible Consumption and Production	Promotes efficient use of resources through automation, smart control systems, and sustainable electronic-system design.
SDG 13	Climate Action	Supports the development of technology-based solutions for energy efficiency, resource conservation, and environmentally sustainable applications.

CO–POMapping(Day1Focus:CO1, CO2)

CO	POMapped	Strength
CO1	PO1,PO2	High
CO2	PO1,PO5	High
CO3	PO1,PO4,PO5	High
CO4	PO2,PO4,PO5	High
CO5	PO3,PO5,PO6	High
CO6	PO3,PO5,PO9,PO10,PO11	High

4. Learning Domain Distribution (3HModel)

The 360-minute session strictly maintains the prescribed course-level domain distribution of Cognitive 45%, Psychomotor 40%, and Affective 15%.

Domain	Weightage	Time(approx.)	FocusinDay1
Cognitive (Head)	45%	162 minutes	Understanding sensor operation, wave propagation, heat detection and calculation, servomotor operation Arduino timing functions.
Psychomotor (Hand)	40%	144 minutes	Identifying and wiring the Arduino UNO and passive infrared sensor; configuring the Arduino program; observing readings; implementing threshold-based detection; simulation/hardware testing and troubleshooting.
Affective (Heart)	15%	54 minutes	Developing safety awareness, careful wiring, systematic observation, responsible handling of electronic components, accurate measurement practices, proper sensor placement, and safe operation of the Arduino-based obstacle detection system

Bloom's Taxonomy Mapping

Bloom's Level	Activity
Remember	Recall the names of commonly used sensors and actuators.
Understand	Explain the working principles and functions of different sensors and actuators.
Apply	Place and connect virtual sensors and actuators correctly in the Wokwi simulation environment.
Analyze	Identify sensor terminals, actuator terminals, Arduino pins, and the correct orientation of components on-screen.

5. Instructor Preparation (BeforeClass)

Before opening Tinkercad/ Wokwi, the instructor should be ready to explain the basic electrical ideas that students will use throughout the course. The explanations below are intentionally simple and should be connected to the sensor based simulation. Physical components are shown only as visual references; all student work remains simulation-based.

5.1 Passive Infrared Sensor

A PIR sensor also called as motion sensor is an electronic device which senses motion using a pair of pyroelectric sensors to detect heat energy in the surrounding environment.

These two sensors sit beside each other, and when the signal differential between the two sensors changes (suppose if a person enters the room), the sensor will engage. It basically catches movement. It has three terminals namely GND, Vcc and signal pin with 3V regulator, time delay controller, sensitivity controller and BIS001.

GND is considered as the negative pin and is connected to the ground of the system. Vcc basically powers up the pin typically 5V. Signal pin is the output pin.

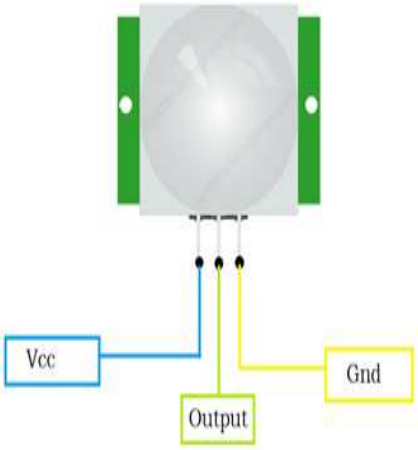


Figure 1: Passive Infrared Sensor

5.2 Servo Motor

The term "servo" comes from *servus*, the Latin word for slave or servant. A servo motor is named this because it acts like a faithful servant to a control system, strictly obeying commands to control position, speed, or accuracy rather than spinning freely on its own.

How Servo Motor Works

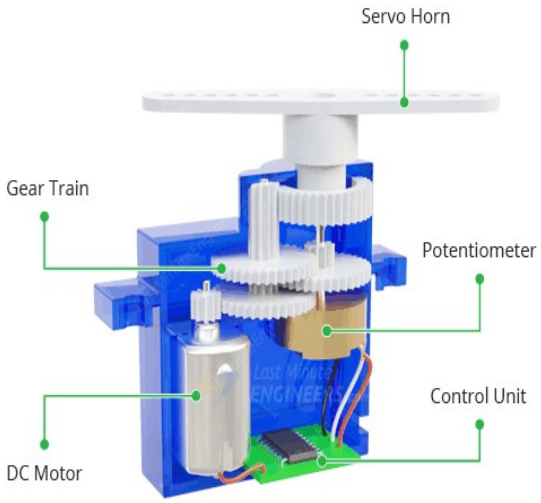


Fig2: Block diagram of the servo motor

Inside a typical hobby servo motor, you'll find five main components working together: a DC motor, a gearbox, a servo horn, a potentiometer, and a control unit.
DC Motor: This is the core component providing the rotational force. It's similar to the hobby motors found in toys, but how it's controlled makes all the difference. The DC motor connects to the output shaft through a series of gears.

Gearbox: The gearbox (a system of gears) plays a crucial role in increasing the motor's torque and improving the precision of its movements. It reduces the speed of the motor but increases its strength.

Servo Horn: This is a plastic arm (or wheel) attached to the output shaft (the part that actually rotates). The servo horn gives you a place to attach whatever you want the servo to move – like the arm of a robot or the rudder on a model airplane.

Potentiometer: This is a variable resistor that acts as a position sensor. It's connected directly to the output shaft. As the output shaft rotates, the potentiometer's resistance changes. This tells the control unit exactly where the motor is positioned.

5.3 Control Unit

Control Unit: This is the “brain” of the servo motor. It receives signals from an external controller (like an Arduino), checks the potentiometer’s position, and controls the DC motor.

Here’s how these parts work together: The servo motor works using what’s called a “closed-loop feedback system“.

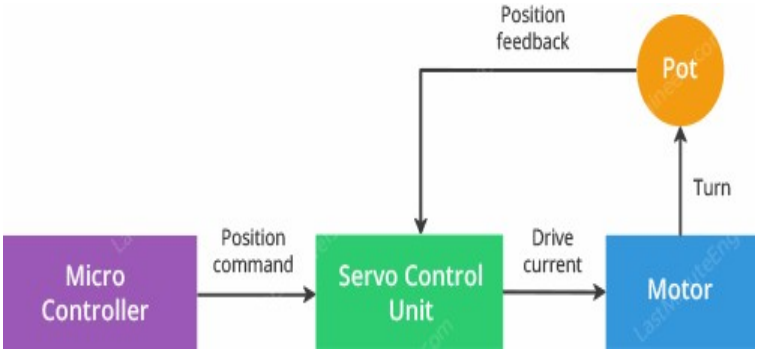


Fig 3: Control block diagram of the servo motor

When a signal is send to the servo telling it to move to a specific position, the control unit receives this command.

The potentiometer is physically connected to the output shaft, so it always knows the current position of the servo. When the motor rotates, the output shaft turns, and so does the potentiometer. This rotation changes the potentiometer’s resistance, creating a voltage that directly corresponds to the current position of the output shaft (and thus the motor). This voltage serves as feedback to the control unit.

The control unit constantly compares this feedback voltage (representing the motor’s current position) with the desired position signal from the microcontroller .If there’s a difference (an “error” signal) between the desired position and the current position, the control unit adjusts the power and direction of the DC motor to fix this difference. As the motor moves, the potentiometer’s feedback voltage changes, continuously updating the control unit about the motor’s current position. Once the motor reaches the desired position you wanted, the error signal becomes zero, and the control unit stops the motor.

This whole process happens very quickly and repeatedly. The servo is always checking its position and making tiny adjustments to stay exactly where it should be. This is why servos are so good at holding their position, even when external forces try to move them.
This entire system is called a “servomechanism” or simply a “servo.” It works as a closed-loop control system, using negative feedback to precisely control the motor’s speed and direction, allowing it to reach and maintain a specific position.

Hobby servo motors are controlled using a clever technique called Pulse Width Modulation (PWM).PWM works by sending a series of electrical pulses to the servo at regular intervals. For most hobby servos, these pulses are sent about 50 times per second (50 Hz). This means a new pulse arrives every 20 ms (1/50th of a second).

5.4 The Buzzer

The piezo, also known as the buzzer, is a component that is used for generating sound. It is a digital component that can be connected to digital outputs, and emits a tone when the output is HIGH. Alternatively, it can be connected to an analog pulse-width modulation output to generate various tones and effects. The **Grove Buzzer** operates at both 3.3V and 5V with a sound output of 85 decibels. This module can be used to provide sound feedback to your application just like the click sound of a button on a digital watch.

Inside the buzzer there is a membrane, like a drumhead, that vibrates due to electric current. When tone(BUZZER) is executed, the current passes through the membrane and deforms it. For noTone(BUZZER) the current stops and the membrane goes back to its original shape. When current is turned on and off very quickly, the membrane vibrates back and forth, creating sound waves that are audible. By changing how fast the current is turned on and off, you can control the speed of the vibration. Faster vibrations create higher pitches while slower vibrations create lower pitches.

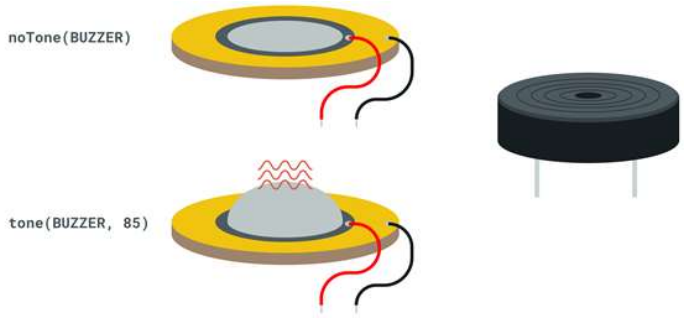


Fig 4: Diagram of the buzzer

5.5 Ohm's Law

Ohm's Law states that the current through an ohmic conductor is directly proportional to the voltage across it and inversely proportional to its resistance, provided temperature and other physical conditions remain constant. The relationship is V = I × R. It may also be rearranged as I = V/R and R = V/I.

Worked example: For a 9 V source and a 1 kΩ resistor, I = 9/1000 = 0.009 A = 9 mA. This simplified calculation treats the resistor alone; in the LED circuit, the LED voltage drop must also be considered :I≈(Vsupply−VLED)/R.

Purpose: To integrate a PIR motion sensor, servo and buzzer to create an automatic robotic response.

Objectives

- Read a digital motion event.
- Apply conditional control.
- Move an actuator automatically.
- Generate an event alert.

No.	Requirement	Quantity	Purpose
1	ArduinoUNO	1	Microcontroller board; acquires the echo and drives all outputs
2	PIRsensor	1	Time-of-flight heat measurement
3	MicroServomotor	1	To provide precise, controlled angular or linear movement in small electronic and robotic projects. Unlike regular motors that spin endlessly, a micro servo moves to an exact angle (usually between 0 and 180 degrees) and holds that position firmly
4	Buzzer	1	To convert electrical energy into an audible sound signal. It serves as a simple and cost-effective indicator to provide alerts, warnings, or confirmation feedback to a user when an event happens, a timer finishes, or an error occurs.
5	Breadboard	1	Circuit assembly
6	Jumper Wires	As required	Interconnections
7	USB Cable / Wokwi account		Programming, power and Serial Monitor access

ArduinoProgram

```
    #include<Servo.h>
    Servogate;
    constintPIR=2,B=6;
      void setup(){
Serial.begin(9600);
pinMode(PIR,INPUT);
pinMode(B,OUTPUT);
gate.attach(9);
gate.write(0);
    }
      void loop(){
if(digitalRead(PIR)){
        Serial.println("MOTIO
  N"); gate.write(90);
  tone(B,1500);
       } else {
  Serial.println("SAFE");
  gate.write(0);
  noTone(B);
    }
    delay(100);
   }
```

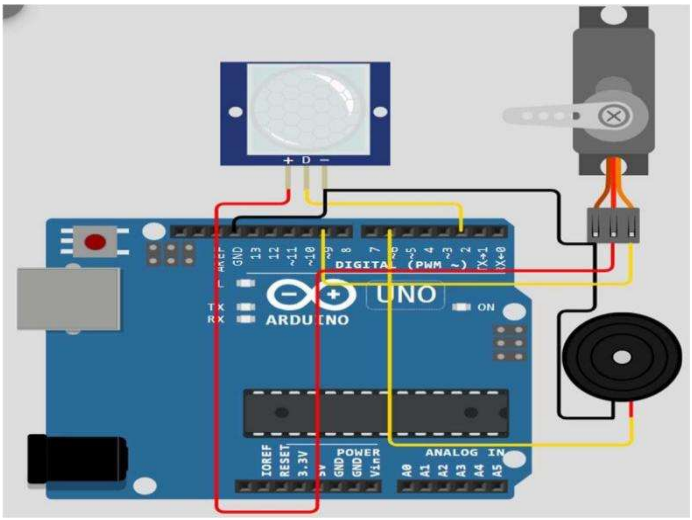


Figure5:.Verified PIR–servo–buzzer automatic response circuit.

6. Software & ResourcesRequired

Resource	Purpose

Tinkercad (www.tinkercad.com)	Primary simulation platform for circuit design, Arduino programming, and temerature sensor-based measurement and servomotor operation with buzzer.
Wokwi (wokwi.com)	Alternative simulation platform for Arduino-based circuit design, programming, and testing of temp sensor .
Web browser(Chrome/Edge, updated)	Tinkercad and Wokwi run directly in the browser; no software installation is required.
Projector/Screen-share	Enables live instructor demonstrations of circuit wiring, programming, sensor operation, temp detection, and servo motor operation.
Observation Sheet(Day1)	Used by students to record signals, measured temp, observations, and experimental outcomes.

This experiment follows a Hardware-Integrated and Simulation-Based approach. Physical components, kits, and Arduino- based hardware are used for hands-on implementation and verification, while Tinkercad /Wokwi are used for circuit design, programming, simulation, and Verified PIR–servo–buzzer automatic response circuit.

7. Student Preparation(Pre-Lab)

- Create a free Tinkercad account prior to the session (or join instructor's Tinkercad Classroom via class code).
- Read the short pre-lab material on temp sensors, Arduino UNO, servomotor, LEDs, and buzzer-based alerts before the session.
- Bring a notebook (physical or digital) for observation recording.

8. Minute-wise Instructor &Student Activity Plan*

Time	Instructor Activity	Student Activity	Domain
0–20 min	Introduce the experiment, aim, objectives, laboratory/simulation practices, safety requirements, and applications of PIR-based motion detection and automatic response systems in security, automation, and robotics.	Listen to the introduction, understand the importance of safe laboratory/simulation practices, and demonstrate readiness to participate in the activity.	Affective (Heart)
20–60 min	Explain the working principle of the PIR sensor, including infrared radiation, motion detection, HIGH/LOW output, and the role of the sensor in triggering an automatic response. Discuss practical applications of PIR-based systems.	Understand the operating principle of the PIR sensor and explain its role in motion detection, security systems, automatic doors, lighting systems, and robotics.	Cognitive (Head)
60–95 min	Explain the Arduino UNO, PIR sensor, servo motor, buzzer, LEDs (if used), digital I/O pins, power connections, and the relationship between PIR output and actuator response.	Identify the components and explain the function of each component in the automatic response circuit.	Cognitive (Head)
95–135 min	Demonstrate identification of Arduino UNO, PIR sensor, servo motor, buzzer, and breadboard. Show VCC, GND, OUT, servo signal, power, and buzzer connections using Tinkercad/Wokwi or hardware.	Identify the components and terminals, arrange the components, build the circuit, and verify the required power, signal, and common-ground connections.	Psychomotor (Hand)
135–180 min	Explain the Arduino program using digitalRead(), servo control functions, buzzer control, conditional statements, and PIR sensor states for detecting motion and activating the automatic response.	Interpret the program statements and explain the sequence: PIR sensing → motion detection → Arduino processing → servo movement → buzzer activation → automatic response.	Cognitive (Head)

180–220 min	Demonstrate program entry/upload and guide students in running the PIR–servo–buzzer automatic response program in simulation or hardware.	Enter/upload the Arduino program, start the simulation or hardware test, introduce motion near the PIR sensor, and verify the servo and buzzer responses.	Psychomotor (Hand)
220–260 min	Guide systematic experimentation by testing different motion conditions and demonstrate the use of the Serial Monitor for observing PIR sensor status and response conditions.	Test the system with different motion/no-motion conditions and record at least five observations along with PIR status, servo position, and buzzer response.	Psychomotor (Hand)
260–302 min	Explain PIR detection range, sensing delay, false triggering, sensor warm-up time, servo movement limitations, buzzer response, and common troubleshooting conditions.	Analyze the effect of motion and sensor positioning on system response, compare expected and observed behavior, and identify possible causes of errors involving wiring, sensor orientation, code, power supply, or environmental conditions.	Cognitive (Head)
302–326 min	Guide students in modifying PIR response conditions, servo angles, buzzer duration, and detection logic, followed by testing of different automatic-response conditions.	Modify the program, test different servo positions and buzzer timings, verify the automatic response, and correct simple wiring or programming errors.	Psychomotor (Hand)
326–360 min	Discuss safe operation, responsible handling of electronic components, teamwork, and practical applications of PIR-based automatic systems. Conduct review/Q&A and provide post-lab instructions.	Participate in discussion and Q&A, demonstrate safety awareness and teamwork, summarize the experiment, and note the post-lab work.	Affective (Heart)

9. Instructor Demonstration Steps (Tinkercad only)

Step	Procedure
Step 1	Sign in to Tinkercad and create a new Arduino circuit/project.
Step 2	Add an Arduino UNO, PIR motion sensor, servo motor, buzzer, breadboard, and jumper wires to the workspace.
Step 3	Connect the PIR sensor to 5V and GND, and connect its OUT pin to Arduino digital pin D2. Connect the servo signal pin to D9, with the servo power and ground connected appropriately. Connect the buzzer to D7 and ensure a common ground is maintained.
Step 4	Enter the Arduino program to read the PIR sensor output using `digitalRead()`. Program the Arduino to activate the servo motor and buzzer when motion is detected and return the servo to its initial position when no motion is detected.
Step 5	Start the simulation and introduce motion within the sensing range of the PIR sensor. Observe how the PIR output, servo movement, and buzzer response change between motion and no-motion conditions.
Step 6	Open the Serial Monitor and observe the displayed PIR sensor status and corresponding automatic-response messages such as MOTION DETECTED or NO MOTION.
Step 7	Test the automatic response by moving an object/person within the PIR detection range. Verify that the servo rotates to the programmed position and the buzzer activates when motion is detected, and that the system returns to its normal state when motion stops.
Step 8	Stop the simulation, verify the circuit connections and program, save the project, and rename it as PIR_Servo_Buzzer_Automatic_Response_[Instructor Name].
Step 9	Repeat Steps 2–8 using hardware components for the physical implementation and verification of the experiment.

10. Guided/Probing Questions

- What is a sensor?
- What does a sensor detect or measure?
- Can you name some sensors you see in everyday life?
- What physical quantity does a temperature sensor measure?
- What happens when a light sensor detects a change in light?
- What would happen if a sensor gives an incorrect reading?
- How would you choose a suitable sensor for measuring water level?
- What is the relationship between a sensor, controller, and actuator?
- Can a system work with an actuator but no sensor? What about a sensor but no actuator?
- How do sensors and actuators work together in a closed-loop control system?
- What would happen if the sensor's output were delayed?
- Can you think of a real-world system that uses multiple sensors and actuators?

11. Hands-on Activity(StudentTask)

Task: Build and simulate a PIR–Servo–Buzzer automatic response circuit using Arduino in Tinkercad/Wokwi.

- Log in to your Tinkercad/Wokwi account or join the instructor's classroom.
- Create a new circuit named “PIR_Servo_Buzzer_[YourName]_[RollNo]”.
- Place an Arduino UNO, PIR sensor, micro servo motor, and buzzer in the simulation workspace.
- Connect the PIR sensor to the Arduino and configure it to detect motion.
- Connect the servo motor to the Arduino and configure it as the actuator.
- Connect the buzzer to the Arduino to provide an audible alert when motion is detected.
- Write/upload the Arduino program to read the PIR sensor and control the servo and buzzer automatically.
- Run the simulation and trigger the PIR sensor to verify the automatic response.
- Observe and record the servo position and buzzer state for both motion detected and no motion conditions.
- Save the completed circuit and keep the simulation/share link ready for instructor review.

12. Discussion Points

- **PIR Sensor Applications in Automation and Robotics** – Discuss the use of PIR sensors for motion detection and automatic response in security systems, smart lighting, automatic doors, surveillance systems, robotic systems, and home automation.
- **Motion-Based Automatic Response Concept** – Discuss how the PIR sensor detects changes in infrared radiation caused by human or object movement. The Arduino reads the PIR output and processes it to activate the **servo motor and buzzer** when motion is detected, while returning the system to its normal state when no motion is detected.
- **Simulation, Hardware Integration, and Safe Operation** – Discuss the use of **Tinkercad/Wokwi** for circuit simulation and testing, proper PIR sensor, servo, and buzzer connections, systematic testing under motion and no-motion conditions, precautions while handling electronic hardware, troubleshooting, and organized project management.

13. Reflection Questions(End of Session)

What is one new concept you learned about the PIR–Servo–Buzzer automatic response system?
- Which part of the circuit did you find easiest to understand, and which part was confusing?
- How confident do you feel about connecting and testing the PIR sensor, servo motor, and buzzer? Rate yourself from **1–5** and explain why.
- What improvement would you like to make to this automatic response system in a future activity?

14. Home Assignment

On Tinkercad/Wokwi, identify and list three different sensors and three different actuators and write one application of each.
- Modify the PIR–Servo–Buzzer circuit so that the servo moves to 180° when motion is detected and returns to0° when no motion is detected.
- Modify the program so that the buzzer produces a different alert response when motion is detected, and observe the change during simulation.

- Write a short note explaining how the PIR sensor, Arduino, servo motor, and buzzer work together to create an automatic response system.

15. Assessment Rubrics

Rating Scale: 4 = Excellent, 3 = Good, 2 = Satisfactory, 1 = Needs Improvement (aligned to course domain weightage: Cognitive 45% / Psychomotor 40% / Affective 15%)

Criterion	Domain	4 – Excellent	3 –Good	2 – Satisfactory	1 – Needs Improvement
Sensor and Actuator Identification and Classification	Cognitive	Correctly identifies and classifies all sensors and actuators and clearly explains their functions.	Correctly identifies and classifies most sensors and actuators with suitable explanation.	Identifies some sensors and actuators but needs support in classification and explanation.	Unable to correctly identify or classify the sensors and actuators.
Sensor and Actuator Interfacing in Tinkercad	Psychomotor	Independently connects sensors and actuators with correct power, ground, signal connections, and polarity; the system functions properly.	Constructs a functional sensor–actuator circuit with only minor errors or limited instructor assistance.	Completes the circuit with considerable instructor assistance; some connection errors remain.	Circuit is incomplete, incorrectly connected, or non-functional.
Experimentation with Sensor Inputs and Actuator Outputs	Psychomotor	Tests different sensor inputs, records observations accurately, and clearly explains the corresponding changes in actuator output.	Tests different sensor inputs and records most observations correctly.	Tests limited sensor inputs and records incomplete observations.	Does not perform or record the sensor–actuator experiment.
Laboratory Engagement and Teamwork	Affective	Participates actively, follows virtual laboratory practices, and consistently supports peers.	Participates regularly and cooperates well with peers.	Participates only when prompted and shows limited teamwork.	Remains passive, disengaged, or does not cooperate with peers.

16. Student Observation Sheet(Template)

S.No.	Motion Condition	PIR Sensor Output	Servo Position	Buzzer State	System Response
1					
2					
3					
4					

Expected Output / Decision Logic

Motion Condition	Indicator	Expected Behaviour
No Motion	NORMAL	Servo remains at initial position + Buzzer OFF
Motion Detected	ALERT	Servo rotates to programmed position + Buzzer ON
Motion Continues	ALERT	Servo remains at programmed position + Buzzer continues as programmed
Motion Stops	NORMAL	Servo returns to initial position + Buzzer OFF

17. Expected Learning Outcome

At the end of the session, a student should be able to independently construct and simulate a PIR–Servo–Buzzer automatic response circuit using Arduino, correctly interface the PIR sensor with the servo motor and buzzer, run and test the simulation for motion and no-motion conditions, and explain how sensor input is processed to produce an automatic actuator and alert response.

18. Instructor Tips

- Keep the **first 25 minutes strictly conceptual and crisp**, covering only the fundamentals of sensors, actuators, inputs, outputs, and their industrial applications. Ensure that the **majority of the session (at least 60%) remains hands-on in Tinkercad**.
- During the hands-on segment **(70–95 min)**, circulate constantly—physically in the classroom or through screen-share support. Pay particular attention to **wire connection points, component terminals, and circuit continuity**, as these are common challenges for first-time Tinkercad users.
- Encourage **peer-assisted learning** during the **95–108 min discussion window**. Pair faster learners with students who need additional support so that students can explain and troubleshoot circuits collaboratively.
- **Do not introduce programming or Arduino coding** at this stage. The session should focus exclusively on **sensors, actuators, basic circuits, wiring, and simulation in Tinkercad**.
- Use **positive and encouraging language** when students make wiring or simulation errors. Treat mistakes as part of the learning process; this session should establish a supportive learning environment and positive attitude toward troubleshooting.
- **Do not introduce or reference physical hardware, kits, breadboards, or other physical components.** The course is delivered entirely through **Tinkercad and the TI software suite**, so all demonstrations, activities, and troubleshooting should remain within the software environment.

CSE 4571 —PRACTICAL ROBOTICS PROJECTS WITH ARDUINO

BasicQuestionsfortheDay1Session

No.	Question	Learning Domain	TaxonomyLevel
1	What is a sensor?	Cognitive	Bloom's Taxonomy – Remember
2	What is an actuator?	Cognitive	Bloom's Taxonomy – Remember
3	What is a PIR sensor?	Cognitive	Bloom's Taxonomy – Remember
4	What is the function of a PIR sensor in an automatic control system?	Cognitive	Bloom's Taxonomy – Understand
5	What is a servo motor?	Cognitive	Bloom's Taxonomy – Remember
6	What is the function of a buzzer in an automatic response circuit?	Cognitive	Bloom's Taxonomy – Understand
7	What type of movement can a servo motor provide?	Cognitive	Bloom's Taxonomy – Remember
8	How does a PIR sensor detect human movement?	Cognitive	Bloom's Taxonomy – Understand
9	What happens when the PIR sensor detects motion in the circuit?	Cognitive	Bloom's Taxonomy – Understand
10	Why is a servo motor used as an actuator in this system?	Cognitive	Bloom's Taxonomy – Understand
11	How does the buzzer provide an automatic response to motion?	Cognitive	Bloom's Taxonomy – Understand
12	What will happen if the PIR sensor detects no movement?	Cognitive	Bloom's Taxonomy – Analyze
13	How can a PIR sensor, servo motor and buzzer be combined to create an automatic response system?	Cognitive	Bloom's Taxonomy – Apply
14	What would happen to the system if the PIR sensor output were disconnected from the controller?	Cognitive	Bloom's Taxonomy – Analyze
15	Why is a power supply required for the PIR sensor, servo motor and buzzer?	Cognitive	Bloom's Taxonomy – Understand
16	Can you identify the PIR sensor, servo motor, buzzer and controller from the circuit components?	Psychomotor	Simpson's Taxonomy – Perception
17	Can you locate the required PIR sensor, servo motor and buzzer in the Tinkercad component library?	Psychomotor	Simpson's Taxonomy – Perception
18	Can you place the PIR sensor, servo motor and buzzer correctly in the workspace?	Psychomotor	Simpson's Taxonomy – Set
19	Can you identify the input and output terminals of the PIR sensor?	Psychomotor	Simpson's Taxonomy – Perception
20	Can you connect the PIR sensor to the controller with correct wiring?	Psychomotor	Simpson's Taxonomy – Guided Response
21	Can you connect the servo motor to the controller correctly?	Psychomotor	Simpson's Taxonomy – Guided Response
22	Can you connect the buzzer to the circuit with correct polarity and connections?	Psychomotor	Simpson's Taxonomy – Guided Response
23	Can you complete the PIR–Servo–Buzzer automatic response circuit correctly?	Psychomotor	Simpson's Taxonomy – Guided Response
24	Can you start the simulation and verify the response of the servo	Psychomotor	Simpson's Taxonomy –

	motor and buzzer when motion is detected?		Mechanism
25	Can you modify the circuit so that the servo motor changes its position when the PIR sensor detects motion?	Psychomotor	Simpson's Taxonomy – Mechanism
26	Can you test the circuit by changing the motion condition and observe the corresponding actuator response?	Psychomotor	Simpson's Taxonomy – Mechanism
27	Are you willing to follow proper safety practices while constructing and testing the automatic control circuit?	Affective	Krathwohl's Taxonomy – Receiving
28	How actively did you participate and support your peers while developing the sensor-based circuit?	Affective	Krathwohl's Taxonomy – Responding
29	Do you value the use of sensors and actuators for developing automatic control systems? Why?	Affective	Krathwohl's Taxonomy – Valuing
30	Do you appreciate automatic response systems as useful applications of electronics and embedded control? Why?	Affective	Krathwohl's Taxonomy – Valuing

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